Important DFT Properties

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Key Concepts

1) A length-N DFT implies the signal is N-periodic in both its time- and frequency-domain representations.

2) The DFT is linear: the DFT of a weighted sum of signals is the weighted sum of the individual signal DFTs.

3) Shifting a signal in time by m samples multiplies the length-N DFT coefficients by $e^{-j2\pi km/N}$.

4) Multiplying a signal by a complex sinusoid with frequency $2\pi m/N$ shifts the frequency of the length-N DFT coefficients by m.

5) The DFT coefficients for a real-valued signal are conjugate symmetric.

    a) The real part of the DFT coefficients is an even function of frequency.

    b) The imaginary part of the DFT coefficients is an odd function of frequency.

    c) The magnitude of the DFT coefficients is an even function of frequency.

d) The phase of the DFT coefficients is an odd function of frequency.

Video Screenshots

DFT Properties

1) Periodicity. Let $x[n] \xleftrightarrow{DFT;N} X[k]$

a) $X[k]$ is N periodic, i.e., $X[k+N] = X[k]$

- follows from 2π-periodicity of $X(e^{j\omega})$

b) $x[n]$ is N periodic, i.e., $x[n+N] = x[n]$

- sampling in freq $\Rightarrow$ periodicity (replicates) in time

2) Linearity

$$ax[n] + by[n] \xleftrightarrow{DFT;N} aX[k] + bY[k]$$

3) Time Shift.

$$x[n-m] \xleftrightarrow{\text{DFT};N} X[k]e^{-j\frac{2\pi}{N}km}$$

4) Modulation

$$x[n]e^{j\frac{2\pi}{N}mn} \xleftrightarrow{\text{DFT};N} X[k-m]$$

5) Conjugate Symmetry. For $x[n]$ real,

$$X^*[k] = X[-k] = X[N-k]$$

- $\text{Real}\{X[k]\}$: even
- $\text{Imag}\{X[k]\}$: odd
- $|X[k]|$: even
- $\arg\{X[k]\}$: odd

6) Inversion.

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j\frac{2\pi}{N}kn} \quad , \quad X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}$$

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] e^{-j\frac{2\pi}{N}km}\right) e^{j\frac{2\pi}{N}kn}$$

$$= \sum_{m=0}^{N-1} x[m] \left(\frac{1}{N}\sum_{k=0}^{N-1} \left[e^{j\frac{2\pi}{N}(n-m)}\right]^k\right)$$

$$= \sum_{m=0}^{N-1} x[m] \begin{cases} 1 , & n=m \\ 0 , & n \neq m \end{cases}$$

$$= x[n]$$

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